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Hardware

GIGA Display Shield



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Display Camera Feed on GIGA Display Shield with OpenMV

Learn how to connect a OV7675 camera module and send the frames directly to the display, using the OpenMV platform.

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The **GIGA Display Shield** has a camera connector which is extended from the **GIGA R1 WiFi** board. The inexpensive 20 pin **OV7675** camera module can be directly mounted without any other circuitry, and using the OpenMV platform, we can directly display the camera feed on the display, at a framerate of about 15~ FPS.

In this guide, you will learn how to set up your **OV7675** camera module with the GIGA R1 WiFi / GIGA Display Shield, using the OpenMV platform.

 Note that the OpenMV IDE includes the firmware and the script for this guide. There is no need for any other external tools.

Hardware & Software Needed

Arduino GIGA R1 WiFi

Arduino GIGA Display Shield

OV7675 camera module

OpenMV (**v4.1.1** and above).

Hardware Setup

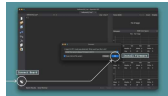
- 1 Mount the GIGA Display Shield on the back side of the GIGA R1 WiFi.
- 2 Mount the OV7675 camera module to the

above long side of the display.

OpenMV Setup

The **OpenMV IDE** includes all tools necessary for flashing your board with the right firmware, as well as the script needed to run the application.

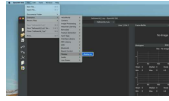
- 1 Download and install the [OpenMV IDE](#).
- 2 Launch the editor, and connect your board to your computer. In the bottom left corner of the editor, click the "Connect" button. The editor should detect your board, and prompt you to install the latest firmware available.



Install the
firmware

- 3 Run the installation, but be aware that this will erase any existing sketch on your board. Wait until it is finished (do not disconnect the board). This can take up to a minute. ***Note that when installing the OpenMV firmware, your GIGA R1 WiFi board will no longer be detected in the Arduino IDE. To go back to "Arduino mode", simply double tap the reset button on the board, and it will be recognized.***
- 4 Once complete, you will need to run a script to see the camera feed. In

Examples > Display > display.py.



Choose the display.py example.

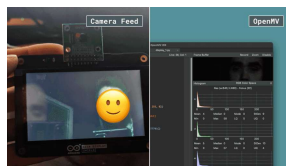
- 5 Click the "Play" button to launch the script. You should now see the camera feed on the GIGA Display Shield.



Click "Play" button to launch script.

Congratulations, you have now set up GIGA R1 WiFi board to stream camera sensor data to the GIGA Display Shield!

The image below demonstrates how the setup should look like when complete.



Final setup.

Code Example

The full script for this guide can be copied from below:

 Code Source on [Github](#)

```
1  # This work is licens
2  # Copyright (c) 2013-
3  # https://github.com/
4  #
5  # LCD, touch panel an
6
7  import sensor
8  import time
9  import image
10 import display
11 from gt911 import GT9
12 from machine import I
13
14 IMG_OFFSET = 80
15 touch_detected = Fals
16 points_colors = ((255
17
18 sensor.reset() # Res
19 sensor.set_pixformat(
20 sensor.set_framesize(
21
22 lcd = display.DSIDisp
23     framesize=display
24 )
25
26 # Note use pin number
27 # driver needs to cha
28 touch = GT911(
```

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